

ACETONE

Synonyms

Dimethyl formaldehyde, dimethyl ketone, 2-propanone

Description

Highly volatile solvent used for varnishes, lacquers, resins, rubber cement, glues, and nail polishes. Primary ingredient in many brands of nail polish removers (acetone-free nail polish removers may use ethyl acetate as the solvent).

SI Unit Conversion

blood acetone (mg/dL) = blood acetone (μmol/L) x 0.0058

Toxicity

Acetone is a product of fat metabolism and is present in very low concentrations in healthy individuals. Diabetic ketoacidosis, fasting, and isopropyl alcohol exposure may result in elevated acetone concentrations. **See ISOPROPYL ALCOHOL.**

Blood acetone levels of 5682 μmol/L (33 mg/dL) have not been associated with symptoms. Blood acetone level of 43,050 μmol/L (250 mg/dL) was recorded in one adult overdose, and 76,630 μmol/L (445 mg/dL) was recorded in a severely toxic pediatric patient.

Mechanism of Toxicity

Not established.

Toxic Dose

Threshold limit value (TLV) is 750 ppm. Ingestion of 2-3 mL/kg may produce severe toxicity. Ingestion of one or two mouthfuls (5-10 mL by a child, 15-30 mL by an adult) is not considered harmful.

Case Reports

Ingestion of 200 mL by an adult produced coma lasting 12 hours.

A 30-month-old child survived ingestion of up to 6 ounces (180 mL) of nail polish remover (maximum 13 mL/kg acetone plus 2 mL/kg isopropyl alcohol). Symptoms included coma, seizures, respiratory depression, hyperglycemia and ketoacidosis. Prolonged dermal exposure during orthopedic cast application has resulted in systemic toxicity in children.

Pharmacokinetics

Well absorbed following ingestion and inhalation, slowly absorbed through skin. Primarily excreted unchanged in urine and via lungs; slowly metabolized to acetate and formate. Metabolism appears to be saturable at high levels of exposure. Children may metabolize acetone more rapidly than adults. Half-life varies with magnitude of exposure. In experimental exposure of adults, an elimination half-life of 5 hours was seen following oral ingestion of 12 mL, and a half-life of 3 hours was noted following short term inhalation exposure at levels below the TLV. Half-lives of 19-31 hours have been reported following overdose.

Clinical Effects

Topical:

Skin contact may result in local irritation, dryness; prolonged contact may result in systemic symptoms. Chronic use of acetone nail polish removers may lead to brittle, peeling, and splitting nails.

Ocular:

A drop splashed in the eye can produce immediate stinging and mild epithelial injury which resolves within a day or two. Prolonged eye contact with a solution containing acetone and cellulose acetate caused permanent corneal opacity. High vapour concentrations may cause eye irritation.

Inhalation:

High concentrations may result in respiratory tract irritation and produce systemic effects.

Ingestion:

General: CNS depression and gastrointestinal symptoms are the most common effects. Onset of symptoms may be delayed. Acetone breath odour may be present following all routes of exposure.

Respiratory: Respiratory depression.

Neurologic: CNS symptoms resemble ethanol intoxication and include ataxia, slurred speech, headache, diplopia, drowsiness. Coma may be prolonged. Seizures are rare.

GI: Nausea, vomiting (onset may be delayed 6-12 hours).

GU: Renal tubular necrosis has been reported rarely.

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Fluid/Lytes/Acid-Base: Ketosis and hyperglycemia (common); may be accompanied by polyuria, polydipsia. Metabolic acidosis may be present. Severe intoxication may resemble diabetic ketoacidosis; however the degree of hyperglycemia is often not as great as would be expected in a diabetic patient.

Lab: May interfere with some serum creatinine assays resulting in falsely elevated creatinine.

Treatment

1. **Topical:** Wash skin thoroughly with soap and water.
2. **Ocular:** Flush eyes with gentle stream of tepid water for 5 minutes. Obtain ophthalmologic opinion if eye irritation persists.
3. **Inhalation:** Remove from exposure. Protect airway and assist ventilation as needed
4. **Ingestion:** Activated charcoal does not readily adsorb acetone but may be useful if product contains other ingredients. If ingested product is a nail polish remover, verify specific formulation as other toxic ingredients may be present.
5. Protect airway and assist ventilation as needed.
6. Monitor blood glucose, blood and urine ketones, electrolytes. *Caution:* acetone may interfere with some laboratory tests including some methods of creatinine determination.
7. Maintain fluid and electrolyte balance.
8. Symptomatic and supportive treatment.